MiniZinc Derby

5 instances · Download all inputs (.zip)

MiniZinc Derby

You are one of the organizers of this year's MiniZinc Challenge. Although many excellent solvers compete in the challenge, watching them process hundreds of benchmark instances is not particularly interactive for conference participants.

You therefore decide to introduce the **MiniZinc Derby**.

The idea is inspired by the traditional CADE ATP System Competition sweepstake, in which conference participants try to predict which prover will perform best on selected problems. Unlike Geoff, however, you are not brave enough to run the entire competition live during the conference.

Fortunately, the MiniZinc Challenge is organized impeccably. All solver results are available well before the conference - unlike in certain other competitions that shall remain unnamed. You can therefore reveal the benchmark results gradually. On each day of the conference, you present several instances and update the cumulative ranking of the competing solvers.

Since the results are already known, you may choose the order in which the instances are revealed. Your goal is to choose an order that makes the Derby as entertaining as possible.

After each revealed instance, the solvers are ranked according to their cumulative scores. Several solvers may be tied for first or last place.

Let (L_i) be the set of solvers in first place after the first (i) instances have been revealed, and let (T_i) be the set of solvers in last place.

For every (i>1): * the leading position changes if (L_i != L_{i-1}); * the trailing position changes if (T_i != T_{i-1}).

A **change of the leading position** contributes **2** entertainment points, while a change

of the **trailing position** contributes **1** entertainment point. A change is counted whenever the corresponding set changes, including cases in which one solver enters or leaves a tied position.

Goal

Determine an **ordering** of the instances that **maximizes** the total number of entertainment points.

Input Format

The input describes a single instance (one track) where each line contains a 3-tuple `I`, `T`, `S` as follows:

- `I` is the instanceID,
- `T` is the teamname, and
- `S` is the score of the instance

Observe that therefore, we can easily compute the number of instances by the number of distinct `I` occurrences. Similarly, for the number of teams, which can be calculated by the number of distinct `T` occurrences in the file.

Output Format

It suffices to output a permutation of instances. We therefore expect that we obtain a single line comprising InstanceIDs of the following form:

```
I I I ... I
```

Example

Instance in input:

```
1 1 10
1 2 20
2 2 50
2 3 20
3 1 5
3 3 15
```

For the example input file above, we expect the following output, which indicates that Instance 3 shall be scheduled before Instance 1, which shall be scheduled before Instance 2:

Solution in output:

```
3 1 2
```

This solution output indeed is optimal.

Instances

Submit all instances at once

Upload one .zip containing a file per instance, each named after its instance number (e.g. `instance.1.out`, `2.txt`).

Browse... No file selected.

Submit all

Your code

Upload a .zip of the code your team used to generate the answers for this problem. Re-uploading replaces the previous file.

Browse... No file selected.

Upload code

Instance 1

▸ Show input

Paste your answer

Paste output here, or upload a file below

or upload a file

Browse... No file selected.

Submit instance 1

Instance 2

▸ Show input

Paste your answer

Paste output here, or upload a file below

or upload a file

Browse... No file selected.

Submit instance 2

Instance 3

▸ Show input

Paste your answer

Paste output here, or upload a file below

or upload a file

Browse... No file selected.

Submit instance 3

Instance 4

▸ Show input

Paste your answer

Paste output here, or upload a file below

or upload a file

Browse... No file selected.

Submit instance 4

Instance 5

▸ Show input

Paste your answer

Paste output here, or upload a file below

or upload a file

Browse... No file selected.

Submit instance 5